
MACROMOLECULAR COMPOUNDS
AND POLYMERIC MATERIALS

Influence of the Peroxide Vulcanization Coagent on the Properties of a Blend of Polypropylene with Hydrogenated Styrene–Butadiene–Styrene Block Copolymer

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Abstract—The dependence of the properties of a polypropylene blend with hydrogenated styrene–butadiene–styrene block copolymer, prepared under the conditions of dynamic vulcanization, on the concentrations of the peroxide and various vulcanization coagents (triethylene glycol dimethacrylate, triallyl cyanurate, triallyl isocyanurate) was studied. The influence of the kind and concentration of the peroxide and coagent on the deformation and strength properties of the thermoplastic vulcanizate obtained was accounted for by the experimental data on variation of the gel fraction content of the mixture, vulcanization network density, and degree of crystallinity and molecular mass of polypropylene. The possibilities and conditions for reducing the compression set and enhancing the oil resistance of the thermoelastoplastic were determined.

Keywords: polypropylene, styrene–butadiene–styrene block copolymer, thermoplastic vulcanizate, peroxide, vulcanization coagents

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Thermoelastoplastics (TEPs) exhibiting vulcanized rubber properties can be repeatedly processed into various items by highly efficient methods used for thermoplastics, which favors advanced development of the market of these materials [1, 2]. The most widely used TEP materials are those based on styrene–butadiene–styrene (SBS) ternary block copolymers on hydrogenated SBS (HSBS), and on their blends with other polymers, mainly polypropylene (PP) [3–6]. However, in the compression set and resistance to hydrocarbon media these TEPs are considerably inferior not only to vulcanized rubbers, but also to dynamically vulcanized TEPs based on polypropylene–rubber blends [1, 2]. This fact considerably restricts the application field of TEPs based on SBS and HSBS and makes urgent the problem of reducing their compression set.

The most apparent way to reduce the compression set and enhance the solvent resistance of various elastomer materials is their vulcanization. Therefore, the problem can be solved by vulcanization of a blend of a styrene block copolymer with a thermoplastic in

the course of compounding (dynamic vulcanization) [7, 8]. Published data on vulcanization of blends of thermoplastics with styrene block copolymers are extremely scarce, and data on the effect of various vulcanization agents on the structure and properties of such thermoplastic vulcanizates (TPVs) are lacking or limited [8]. Therefore, it was interesting to study how various peroxide vulcanization coagents influence the formation, structure, and properties of a thermoplastic vulcanizate based on PP and HSBS.

EXPERIMENTAL

As investigation objects we chose polypropylene of PP8300G grade (PAO Nizhnekamskneftekhim), which was a propylene–ethylene block copolymer with the melt flow index (MFI) of 1.2 g/10 min and bending elastic modulus of 1300 MPa, and hydrogenated styrene–butadiene–styrene copolymer of SeptonV9461 grade (Kurarey, Japan), containing 30 wt % bound styrene; the viscosity of its 5% solution in toluene was 0.09 Pa s.

White paraffin oil of Primol382 grade (ExxonMobil) with the kinematic viscosity at 100°C of 8.9 mm² s⁻¹ was used as a plasticizer.

Vulcanization was effected with 2,5-dimethyl-2,5-di(*tert*-butylperoxy)hexane (DTBPH, trade name Trigonox 101, AkzoNobel). As vulcanization coagents we used triethylene glycol dimethacrylate (TGM), triallyl cyanurate (TAC), and triallyl isocyanurate (TAIC), produced by the Kargin Research Institute of Polymers.

Experiments were performed with a blend with a constant PP : HSBS : oil ratio of 30 : 70 : 70. All the blend samples contained antioxidants: 0.9% Irganox 1010 and 0.2% Irgafos 168 (BASF).

Compounding and vulcanization of the components were performed in a Brabender laboratory mixer with the chamber volume of 60 mL at a rotor rotation rate of 150 rpm and a temperature of 170°C. The components were loaded in the following sequence: polypropylene, antioxidant, HSBS, oil, vulcanization coagent, and peroxide.

The vulcanizate of the oil-filled HSBS was prepared as follows. The components (HSBS powder, oil, vulcanization coagent, and peroxide, HSBS : oil ratio 1 : 1) were compounded at room temperature. The resulting compound was vulcanized under a press in a mold at 180°C for 5 min.

The nominal tensile strength σ and relative elongation at break ϵ were measured on a 2001P-0,5 tensile-testing machine in accordance with ISO37 at a clamp velocity of 500 mm min⁻¹. The Shore hardness A was measured according to ISO 7619 with an IT5078 hardness meter at 23 ± 2°C.

The relative static compression set (CS) was measured according to ISO815 by method A after keeping for 22 h at 70°C. Pieces in the form of cylinders 13 mm in diameter and 6 mm high for CS measurements were prepared by pressing.

The TEP resistance to hydrocarbons was evaluated by the degree of swelling α in N3 oil. The test was performed according to GOST (State Standard) ISO 1817–2016 for 163 h at 125°C.

The sol–gel analysis of TEP was performed by extraction of the soluble fraction with refluxing *o*-xylene for 16 h in a Soxhlet apparatus [8, 9]. The insoluble residue was dried at 80°C for 24 h and after drying was

considered as the gel fraction. The relative content of the gel fraction was calculated by the formula

$$\text{Gel fraction} = \frac{m_1}{m_0} \times 100\%,$$

where m_1 is the weight of the dry gel fraction after the extraction (g), and m_0 is the sample weight prior to the extraction (g).

The density ν of the vulcanization network isolated from the TEP gel fraction was determined from the equilibrium swelling in toluene. To this end, the dry gel fraction was placed in toluene and kept there at 20 ± 2°C for 24 h. The vulcanization network density was calculated by the Flory–Rehner equation [10]

$$\nu = -\frac{\rho_2}{\mu_2} \frac{\ln(1-\phi_1) + \phi_1 + \chi\phi_1^2}{\phi_1^{1/3} - \frac{2}{f}\phi_1},$$

where ρ_2 is the solvent density (g cm⁻³); μ_2 , solvent molecular mass (g mol⁻¹); f , functionality of the network chains (= 4); χ , polymer–solvent interaction parameter (0.503 at 20°C) [11]; and ϕ_1 , volume fraction of the elastomer in the swollen sample,

$$\phi_1 = \frac{V_0}{V_1} = \frac{m_3 / 0.91}{m_3 / 0.91 + (m_2 - m_3) / \rho_2},$$

where m_2 is the weight of the dry sample before immersion in toluene (g), m_3 is the weight after swelling in toluene (g), and 0.91 g cm⁻³ is the HSBS density.

The molecular mass (MM) of the initial PP and of PP extracted from TEP was determined viscometrically with an Ubbelohde viscometer [12] at 135°C in a decalin solution. The molecular mass was determined using the Mark–Houwink equation:

$$[\eta] = KMM^\mu,$$

where $[\eta]$ is the intrinsic viscosity (mL g⁻¹), and K and μ are constants for the polypropylene–decalin system, 0.001580 and 0.77 [13], respectively, at 135°C.

The vulcanization rate was estimated indirectly from the rate of the increase in the torque, V_m , on the mixer shaft in the course of dynamic vulcanization. In the curve of the torque vs. time, the portion in which the torque rapidly increased was chosen, and V_m was calculated as

the ratio of the torque increment to the time interval in which it occurred.

The degree of crystallinity X_c of PP in TEP was determined by differential scanning calorimetry (DSC) [14] with a Q-200TA device under nitrogen at a heating rate of 2 deg min⁻¹. Prior to recording DSC, the TEP samples were annealed at 130°C for 1 h to level off the differences in the sample crystal structure, caused by the effect of the temperature and time factors of the sample preparation. When calculating the degree of crystallinity, we took into account the fraction of PP in TEP. The value of 160 J g⁻¹ [14] was taken for the thermal flux in melting of 100% crystalline PP.

RESULTS AND DISCUSSION

In the first step, we studied the dependence of the TEP properties on the DTBPH content (without vulcanization coagent). With an increase in the DTBPH concentration, the deformation and strength characteristics (σ , ϵ , HSA) decrease, and the CS of TEP increases (Table 1). It should be noted that introduction of DTBPH does not cause an increase in the torque on the mixer shaft: V_m is equal to zero (Table 1).

It is known that polypropylene and HSBS are incompatible, and their compounding yields a system with an interpenetrating structure at PP concentrations in the interval 10–55 vol % [3–5]. Therefore, to account for the facts observed (Table 1), we should take into account changes occurring in the phase of each polymeric component under the action of DTBPH.

First, we analyzed changes in the HSBS phase. The sol–gel analysis shows that introduction of 1.6% DTBPH leads to the formation of 17% gel fraction in the mixture, which corresponds to 42% of the weight of HSBS present in TEP. Cross-linking, as a rule, enhances the strength and decreases CS of the HSBS phase. For example, the experiments show that introduction of 1.6% DTBPH increases the strength of oil-filled HSBS (HSBS : oil = 1 : 1) from 1.3 to 1.6 MPa. As a consequence, the strength of the TEP as a whole should increase and its CS should decrease; however, actually the pattern is opposite.

The observed decrease in the mechanical properties and increase in CS of TEP (Table 1) are most probably caused by changes occurring in the polypropylene phase under the action of DTBPH. Experiments show that an increase in the DTBPH concentration causes a signifi-

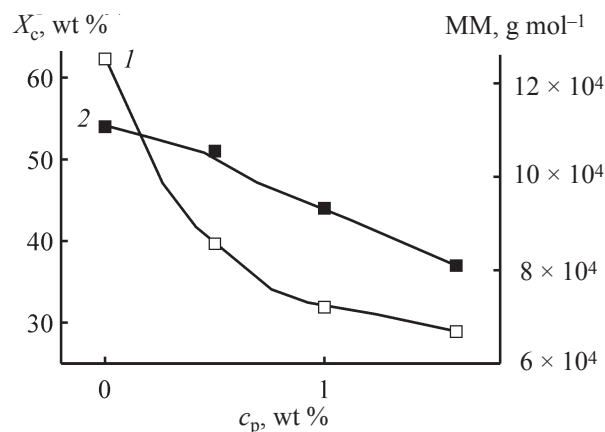


Fig. 1. (1) Molecular mass MM and (2) degree of crystallinity X_c of the polypropylene phase in TEP as functions of the DTBPH content c_p .

cant decrease in the molecular mass MM and degree of crystallinity X_c of the polypropylene phase (Fig. 1). A decrease in MM of polypropylene under the action of radicals is well known; it occurs by the β -scission mechanism via abstraction of a hydrogen atom by a peroxy radical from a tertiary carbon atom in the polymer backbone [15] and is used in industry for producing polypropylene with high MFI (more than 35 g/10 min) [16].

A decrease in X_c (Fig. 1) suggests a decrease in the degree of regularity of PP macromolecules, which may be due to the formation of branched and cross-linked molecular structures in PP under the conditions of a radical chain process. It is known that, when introducing peroxide into pure PP, its degree of crystallinity does not noticeably decrease, and branched and cross-linked structures are formed in extremely small amounts [15]. In our case, the addition of HSBS block copolymer to polypropylene under the conditions of peroxide vulcanization causes the formation of branched structures in the polypropylene phase, which allows us with high probability to speak of grafting of polypropylene macromolecules to HSBS. Similar conclusions on causes of the decrease in the crystallinity of the polypropylene phase in its peroxide-vulcanized blends with rubber are also made by other researchers [17, 18].

A decrease in MM and X_c of the polypropylene phase well accounts for the observed decrease in σ , ϵ , and CS of the blend studied (Table 1). Against the background of the significant decrease in the molecular mass of polypropylene, we even do not observe an increase in the torque on the mixer shaft in the course of the mixture

Table 1. Properties of the thermoelastoplastic with different DTBPH content

DTBPH, wt %	CS, %	σ , MPa	ε , %	HSa, units	V_m , N m s ⁻¹
0	55	8.5	880	56	—
0.5	54	8	780	55	0
1	64	4.4	590	52	0
1.6	67	2.5	410	50	0

Table 2. Properties of TEPs prepared with different amounts of vulcanization coagents at a DTBPH concentration of 1.6 wt TGM, TAIC, TAC

Vulcanization coagent	Coagent content, wt %	CS, %	σ , MPa	ε , %	HSa, units	V_m , N m s ⁻¹	α , %
—	0	67	2.5	410	50	0	—
TGM	1	48	4.5	400	63	0.31	180
TGM	3	40	5.4	360	65	0.3	108
TGM	5	41	6	350	65	0.18	108
TGM	7	40.5	6.5	370	65	0.15	102
TGM	9	40	6.95	462	66	0.1	100
TAIC	1	44	4.25	325	60	0.35	130
TAIC	3	43	4.4	225	62	0.52	88
TAIC	5	38	4.5	150	68	0.55	73
TAIC	9	32	4.6	120	70	0.65	67
TAC	1	40	3.9	260	61	0.38	155
TAC	3	36	4.8	220	67	0.52	130
TAC	5	35.5	5.0	170	68	0.65	90
TAC	9	31	5.4	158	70	0.5	58

vulcanization (Table 1), despite formation of up to 40% gel fraction in the HSBS phase. As seen from the experimental data (Table 1), a decrease in MM and X_c of the polypropylene phase exerts a stronger effect on σ , ε , and CS of TEP than formation of up to 40% cross-linked structures in the HSBS phase does. More significant contribution of the polypropylene phase to the TEP properties is not surprising, because in dynamically vulcanized blends of an elastomer with a thermoplastic the latter usually forms the dispersion medium and the former, the dispersed phase [19–21].

Vulcanization of a blend of polypropylene and HSBS solely with DTBPH without coagent does not al-

low reaching the desired practical result. The efficiency of the peroxide vulcanization of the elastomers can be significantly improved by using various organic compounds with several unsaturated bonds; such compounds are commonly termed vulcanization coagents [22, 23]. Then we studied the dependences of the TEP properties on the content of peroxide vulcanization coagents at a constant peroxide concentration (Table 2).

An increase in the concentration of any of the coagents leads to a decrease in CS, ε , and α and to an increase in σ and HSa; the extent of variation of these properties depends on the coagent type (Table 2). TAC causes the most pronounced decrease in CS and α and increase in

the vulcanization rate V_m ; the effect of TAIC is somewhat weaker; TGM causes the least pronounced decrease in CS and ϵ but the most pronounced increase in σ ; TAIC causes the least pronounced increase in σ but the most pronounced decrease in ϵ ; TAC is intermediate between TGM and TAIC with respect to an increase in σ and a decrease in ϵ .

The mechanical properties of the blend studied are determined by parameters of each of the polymeric phases such as the molecular mass and degree of crystallinity of the PP phase, thickness of the HSBS vulcanization network, and mixture morphology.

The experiments show that introduction of any coagent causes an increase in the gel fraction amount and vulcanization network density (Table 3). The most pronounced increases in the gel fraction content, vulcanization network density, and rate of the growth of the torque on the mixer shaft are observed on introducing TAC; with TAIC, the effect is slightly weaker. Introduction of TGM leads to an appreciably lower amount of the gel fraction, to the less dense network, and to lower vulcanization rate. Such effect of the coagent kind on the gel fraction content and network density is quite understandable: TAC and TAIC contain three active groups in the molecule, whereas TGM contains only two active groups. It can be noted that, at relatively high coagent concentrations (5–9%), the gel fraction content exceeds the total content of HSBS, coagent, and peroxide (Table 2). This fact indicates that a certain part of the gel fraction is formed by PP and that polypropylene macromolecules participate in the cross-linking and grafting.

An increase in the gel fraction content and in the vulcanization network density accounts for a decrease in CS and α . Relatively good correlation is observed: The higher the gel fraction content and the network density are (Table 3), the lower is CS of TEP (Table 2) irrespective of the kind of the coagent.

The increase in the gel fraction content and in the density of the vulcanization network of the HSBS phase can also account for an increase in σ and a decrease in ϵ of TEP on introducing the coagents, because the vulcanization enhances the strength of the HSBS phase. For example, an increase in the TAC amount from 0 to 5% in the presence of 1.6% peroxide leads to an increase in σ of the HSBS/oil blend (1/1) without polypropylene from 1.6 to 3.35 MPa and to a decrease in ϵ from 690 to 530%. However, the correlation between the gel

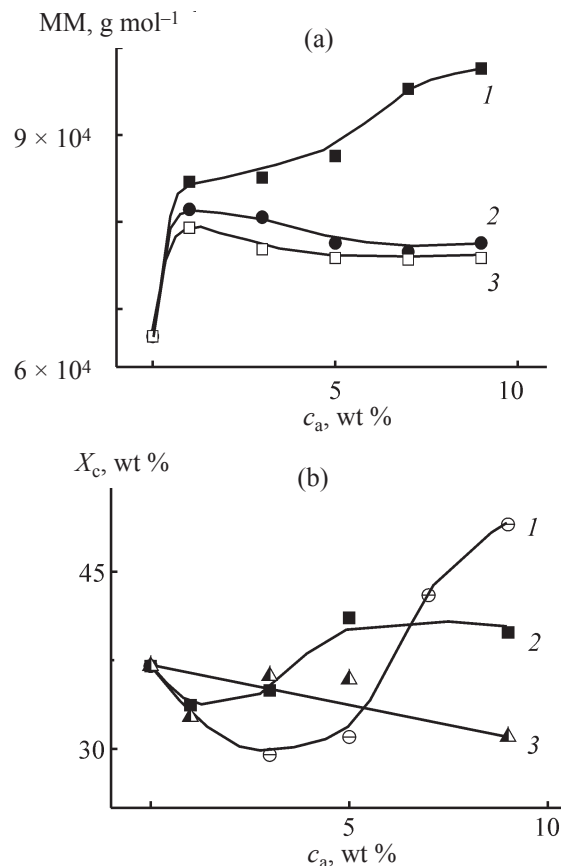


Fig. 2. (a) Molecular mass MM and (b) degree of crystallinity X_c of the polypropylene phase in TEP as functions of the vulcanization coagent content c_a (DTBPH content 1.6 wt %). (1) TGM, (2) TAC, and (3) TAIC.

fraction content and σ of TEP is observed only for each vulcanization coagent separately, with no common correlation for all the coagents (Tables 2, 3). This fact suggests that the strength and relative elongation of TEP are influenced by changes in the polypropylene phase, caused by the coagent introduction.

Therefore, we then studied how MM and X_c of the polypropylene phase in TEP depend on the kind and concentration of the vulcanization coagent at constant peroxide content (Figs. 2a, 2b).

The concentrations of coagents exert complex influence on MM and X_c of the polypropylene phase (Fig. 2), suggesting the diversity of competing chemical processes occurring in the course of vulcanization.

At low concentrations (up to 1%), all the coagents increase MM of polypropylene extracted from TEP. At a higher TGM concentration, MM continues to increase,

Table 3. Content of the gel fraction and density of its vulcanization network in TEP prepared with different amounts of the coagent (DTBPH concentration 1.6%)

Vulcanization coagent	Coagent content, wt %	HSBS, %	Gel fraction, %	$\nu \times 10^4$, mol cm ⁻³
–	0	40	17	0.1
TGM	1	40	35	0.2
TGM	3	39	37	0.25
TGM	5	38	38	0.3
TGM	7	37	44	1.75
TGM	9	36	43	2.35
TAIC	1	40	39	0.15
TAIC	3	39	41	0.4
TAIC	5	38	47	0.8
TAIC	9	36	50	4.8
TAC	1	40	41	0.3
TAC	3	39	45	0.85
TAC	5	38	48	1.4
TAC	9	36	51	5.3

but less efficiently than at low concentration, whereas TAIC and TAC at their increased concentrations cause not a further increase but a slight decrease in MM of the polypropylene phase (Fig. 2).

The increase in MM of polypropylene (Fig. 2a) on introducing peroxide vulcanization coagents was noted by many researchers [15, 24, 25]; it can be accounted for by two factors. Coagents react with polypropylene macroradicals and prevent their scission occurring by the β -scission mechanism [11, 17]. In addition, acrylate and allyl compounds more readily transfer their hydrogen to radicals [23]; therefore, they react with peroxy radicals faster than polypropylene does, thus inhibiting the formation of polypropylene radicals and their subsequent scission. As a result, both these reactions decrease the number of acts of polypropylene macromolecule cleavage.

On the other hand, allyl groups are oxidized more readily than alkanes or aryls [23, 26]. Oxidation of unsaturated compounds with atmospheric oxygen under the conditions of the occurrence of radical chain processes at temperatures higher than 180°C yields peroxy compounds [27], which initiate new

radical chains and cause additional degradation of polypropylene. Therefore, TAC and TAIC at their high concentrations cause a decrease in MM of polypropylene. Similar reversion of the antioxidant action on polymers is also characteristic of many amine and phenol inhibitors of radical processes; they stabilize polymers at low concentrations but are oxidized to peroxides and participate in degenerate branching of radical chains at high concentrations [27].

The dependence of the degree of crystallinity of polypropylene, X_c , on the coagent concentration is also complex. Introduction of a small amount (up to 2–3%) of any coagent causes a decrease in the degree of crystallinity of the polypropylene phase (Fig. 2), which is quite natural and is caused by grafting to polypropylene macromolecules both of the coagents themselves and of polypropylene and HSBS chains, including grafting via coagent. Such action of DTBPH in the presence of coagents on the polypropylene crystallinity was noted by many researchers [8, 17, 19, 28]. Grafting disturbs the regularity of the polypropylene macromolecules and decreases their ability for crystallization. However, at higher TGM and TAC concentrations X_c starts to increase

(to a greater extent in the case of TGM, Fig. 2b). This fact suggests a decrease in the number of acts of grafting to polypropylene macromolecules. Similar dependence (with a minimum) of the degree of crystallinity of the polypropylene phase on the coagent concentration was observed previously for a blend of polypropylene with butadiene–acrylonitrile rubber in vulcanization with peroxide in the presence of TGM [29] and for a blend of polypropylene with HSBS in vulcanization with peroxide in the presence of trimethylolpropane triacrylate [8].

In peroxide vulcanization of polymers, the coagent participates mainly in two competing chemical processes: grafting to the polymer and homopolymerization of the coagent [22, 23, 30]. The results obtained (Fig. 2b) show that such coagents as TGM and TAC in a polypropylene medium at concentrations higher than 2–3% show stronger tendency to homopolymerization than to grafting (copolymerization). Apparently, at a low coagent concentration the probability of collisions of its molecules with each other is low, and the molecules are forced to undergo grafting to the surrounding polypropylene macromolecules. With an increase in the coagent concentrations, the collisions between its molecules become more probable; therefore, the coagent homopolymerization becomes more preferable. This is favored by the fact that the coagents more readily transfer their hydrogen atoms and hence faster react with radicals than polypropylene macromolecules do. However, for TAIC further decrease in X_c is observed even at concentrations higher than 3%, suggesting an increase in the number of acts of grafting to polypropylene (Fig. 2).

However, in the HSBS phase the pattern is different. An increase in the concentration of the coagents throughout the examined interval leads to an increase in the degree of their grafting to macromolecules, as indicated by an increase in the gel fraction content and in the vulcanization network density in this fraction (Table 3).

The observed changes in MM and X_c of the polypropylene phase (Fig. 2) are reflected in the mechanical properties of TEP. The experimentally observed very significant increase in the TEP strength on introducing small (up to 1%) amounts of coagents (Table 2) is due not only to vulcanization of the HSBS phase but also, to a significant extent, to an increase in the molecular mass of polypropylene.

At higher concentrations of the coagents, variation of MM and X_c of the polypropylene phase also influences the mechanical properties of TEP. Owing to the most pronounced increase in the degree of crystallinity and molecular mass of the polypropylene phase on introducing TGM, TGM increases σ of TEP to a greater extent than the other coagents do (Table 2). For the same reason, TEP prepared with TGM is characterized by the highest ϵ . TAIC causes the most pronounced decrease in the degree of crystallinity of the polypropylene phase at the least pronounced increase in the molecular mass (Fig. 2); as a result, with TAIC the increase in σ of TEP is the least pronounced and the decrease in ϵ is the most pronounced (Table 2) among all the coagents. The changes in MM and X_c of the polypropylene phase, caused by the introduction of TAC, are intermediate between those caused by TAIC and TGM (Fig. 2); as a result, the increase in σ and the decrease in ϵ of TEP are also intermediate (Table 2).

CONCLUSIONS

Peroxide vulcanization of the polypropylene–HSBS–oil blend under the compounding conditions in the presence of coagents such as triethylene glycol dimethacrylate, triallyl cyanurate, and triallyl isocyanurate allows the static compression set of the thermoplastic obtained to be considerably decreased and its resistance to hydrocarbons to be enhanced with preservation of satisfactory physicommechanical properties. The nature and concentration of the coagent significantly influence the density of the vulcanization network of the HSBS phase, the degree of crystallinity and molecular mass of the polypropylene phase, and, as a result, the properties of the thermoelastoplastic. The optimum set of properties is reached with triallyl cyanurate used as a coagent.

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